A Quantitative Framework for Portfolio Resilience: Insights from Indian Markets during COVID-19 and Demonetization

~Ananya Arun Panayappan

ABSTRACT

Rare, unpredictable shocks continue to challenge the foundations of financial risk assessment. Black swan events – characterized by correlation breakdowns, heightened volatility, and structural market shifts – render traditional risk metrics inadequate.

To address this, this study develops a Resilience Index integrating volatility, Value at Risk (VaR), correlation difference magnitude, recovery days, and recovery ratio. The index was applied to evaluate sectoral resilience during two disruptions: COVID-19 (2020) and demonetization in India (2016). Subsequent analysis involved portfolio optimization using both the Resilience Index and the Sharpe ratio, with phase-specific weights. The study further derives a generalized form of the Resilience Index, which is validated on multiple black swan events.

The analysis reveals that during COVID-19, Metal was the most resilient sector and Pharma was the least, whereas during demonetization, Media was the most resilient sector and Realty was the least. Importantly, portfolios optimized for both the Resilience Index and the Sharpe ratio achieved better risk-adjusted returns than those optimized solely for the Sharpe ratio in all phases of both events. The generalized Resilience Index demonstrated similar performance during other events.

This framework is a valuable tool to strengthen portfolios during extreme market conditions and provides a foundation for resilience-based asset selection across different markets and event types. These findings are relevant to both portfolio managers and financial researchers.

Keywords: black swan events, Resilience Index, portfolio optimization, sectoral resilience

I. INTRODUCTION

Black swan events are rare events that have a profound impact on the stock market. Examples of black swan events include the global financial crisis in 2008, the COVID-19 pandemic in 2020, demonetization in India in 2016, and the Chinese market crash in 2015. These black swan events have characteristics like correlation breakdown, high volatility, and a shift in market structure. Traditional risk metrics like volatility and Value at Risk (VaR) are designed for normal conditions and therefore fail to capture market behavior during such black swan events. Hence, there is a critical need to measure resilience, which includes the risk during the black swan event and recovery after.

To address this gap, this study introduces a new Resilience Index that utilizes a weighted combination of volatility, Value at Risk, correlation difference magnitude, recovery days, and recovery ratio to capture the impact during a crisis and the speed of recovery after. Weights were calibrated through economic reasoning and heuristics.

Diversified portfolios were constructed using multiple sectoral indices listed in the Indian stock market, including FMCG, Infrastructure, Pharma, Auto, Media, Realty, Energy, Public Sector Enterprises, and PSU Bank. The price paths of these portfolios before, during, and after the black swan events were computed using historical data. This, coupled with Monte Carlo simulation, allowed for the generation of multiple portfolios. By simulating the risks and returns for many portfolios, efficient frontiers were produced for before, during, and after two major black swan events: the COVID-19 pandemic (global, high impact) and the 2016 demonetization in India (local, medium impact). These efficient frontiers were created once to optimize for Sharpe ratios and once to optimize for the weighted sum of resilience and Sharpe ratios. In the

second case, differential weighting specific to the phase of the black swan event was used. The time-period for analysis of the black swan events was selected based on the characteristics of the event. Prolonged events like COVID-19 were analyzed with a 6-month time window for each phase, while a 3-month time window was used for all the phases of demonetization. The time-period selection was optimized to capture short-term market shocks. Additional tests were conducted with event periods selected based on the Indian volatility index.

This study then extends the Resilience Index to generalize it for other black swan events. The generalized Resilience Index was computed by creating a combined Resilience Index that excludes event-specific metrics like recovery days and recovery ratio and finds a geometric mean of the resilience indices during COVID-19 and demonetization. This approach ensured that the generalized Resilience Index gave higher weightage to assets that performed well across both black swan events. Assets that performed well in just one event received lower scores.

The combined Resilience Index was back-tested during different black swan events, like the Chinese market crash, the Russia-Ukraine war, and the GST rollout, which had different characteristics. The Resilience Index demonstrated predictive power and optimized efficient frontiers by reducing risk and increasing returns before, during, and after the events. Further research may be conducted to predict event-specific features like recovery days and recovery ratio based on data available on past black swan events. This research would be of interest to practitioners and investors who will need to optimize for portfolio performance by gaining a deep understanding of short-term market shifts due to black swan events. Researchers may be interested in building upon this framework by adding new features, computing the Resilience

Index for other asset types, and validating the utility of the Resilience Index across a variety of black swan events in different markets and different time periods.

II. LITERATURE SURVEY

This section reviews prior research about black swan events, their impact on financial markets, and approaches to measuring resilience. Tracing the conceptual foundations, empirical evidence, and methodological innovations reveals the gaps in the current literature and establishes the motivation for this study.

A. Conceptual Discussions of Black Swan Events

Early scholars defined black swan events as rare, unforeseeable events that traditional risk management tools fail to capture. Aven (2013) and Flage and Aven (2015) emphasized the need to extend risk management tools since conventional probabilistic methods overlook the uncertainty surrounding extreme events. Building on this foundation, Walter (2020) questioned whether financial black swan events are truly random events, suggesting that better financial models can partially predict such events. This debate reveals a conceptual tension: whether black swan events are truly unpredictable, or whether evolving risk tools can partially anticipate them.

B. Empirical Studies on Sectoral Impacts of COVID-19 and Demonetization

Empirical studies discuss the effects of sudden shocks like COVID-19 and Indian demonetization on the market, highlighting the sharp yet varied effects experienced across sectors.

COVID-19

The literature highlights that the COVID-19 pandemic triggered heightened volatility and sharp declines in global stock markets (Baker et al., 2020). In India, investor sentiment played a critical role in the sharp decline of the stock markets, with lockdowns, policy responses, and mobility restrictions increasing losses (Ozili & Arun, 2023). Sector-specific studies revealed that Indian markets saw a sharp decline in March 2020, with sectors like finance, real estate, and banking most affected (Singh & Neog, 2020). Pharmaceuticals displayed mixed resilience (Donadelli et al., 2017), whereas energy consistently underperformed (Aslam at al., 2020). Interestingly, a rebound in trade volumes was observed after COVID-19 due to domestic confidence in the market (Rishika Shankar and Priti Dubey, 2021). This suggests that sectors experienced a heterogeneous impact, with some sectors proving to be more resilient than others.

Demonetization

Reactions to India's demonetization were similarly uneven. Stock market reactions were inconsistent: state-owned firms and banks benefited the most, with public sector banks gaining immediate positive returns (Dharmapala & Khanna, 2017). At the same time, short-term sectoral losses were seen with immediate restabilization, reaching a 7.1% GDP growth later that fiscal year (Sivankutty, 2017). These findings highlight that resilience is highly context-specific, with dependence on market mechanisms and policy environments.

C. Methodological Approaches to Measuring Resilience

While event studies document the propagation of shocks in markets, another parallel strand of literature focuses on defining and quantifying resilience. Mainik, Mitov, and Rüschendorf (2015)

proposed an extreme risk index that captures tail risk relevant to rare events, while H. Rezaei Sou, A. Esfahanipour, and M. Akbarpour Shirazi (2020) used Value at Risk (VaR) and Conditional Value at Risk (CVaR) to measure risk levels. More recent work by Shaik Kaashif Sahil, Dr. Gangineni Dhananjhay, and Dr. Shaik Karim (2024) offered recommendations to investors and policymakers on the resilience of the Indian stock market, while Singhal (2023) incorporated investor psychology to provide a quantitative measure of risk through the Greed and Fear Index. This methodological evolution reflects a shift from purely statistical measures toward multidimensional approaches that incorporate behavioral, policy, and financial perspectives.

D. Summary of Literature Survey

This literature survey reveals three key trends: debate over the predictability of black swan events, evidence of uneven sectoral impacts during COVID-19 and the Indian demonetization, and improvements in methodology of measuring resilience. However, despite these advancements, few studies integrate resilience into dynamic portfolio optimization frameworks. Existing studies focus either on retrospective analysis of sectors or individual indicators of resilience, but few integrate resilience directly into asset selection and portfolio construction. This study addresses this gap by applying a multidimensional resilience index to the Indian stock market and testing its effectiveness on past black swan events.

III. METHODOLOGY

A. Data Collection

Daily historical closing prices were collected for both broad and sectoral Indian indices, including Nifty 50, Nifty 500, BSE Sensex, Nifty Midcap 100, FMCG, Infrastructure, Pharma, Auto, Media, Realty, Energy, Public Sector Enterprise, and PSU Bank. The data was obtained using the Yahoo Finance API for time periods before, during, and after the COVID-19 pandemic, demonetization, and the Chinese market crash. All data analysis, simulation, and visualization were conducted in Python, utilizing libraries such as pandas, numpy, matplotlib, yfinance, and scikit-learn.

B. Event Selection

The study mainly focuses on two black swan events that had a profound impact on the Indian stock market:

- The COVID-19 pandemic
- Demonetization in India

Each of these events was studied with three windows: before the event, during the event, and after the event. Given that COVID-19 was a unique, global, multi-phase event, a 6-month window was used for all three phases. This longer time frame was selected since it includes both the increased volatility during the pandemic and the gradual recovery phase. Given that demonetization was a local, medium-impact event, a 3-month window was used for all three phases. A 3-month period for each phase effectively captures the immediate liquidity shock,

market adjustment, and later normalization to minimize confounding effects with unrelated macroeconomic events.

A third event, the Chinese market crash, was used to back-test the findings of this study using 3-month windows, like for the demonetization. These periods were specifically chosen to analyze market shifts due to black swan events that affect short-term investors. The following are the dates used for this analysis.

Event	Before Window	During Window	After Window
COVID-19	11 Sep 2019 - 10 Mar 2020	11 Mar 2020 - 10 Sep 2020	11 Sep 2020 - 10 Mar 2021
Demonetization	8 Aug 2016 - 7 Nov 2016	8 Nov 2016 - 7 Feb 2017	8 Feb 2017 - 7 May 2017
China market crash	1 Mar 2015 - 11 Jun 2015	12 Jun 2015 - 30 Sep 2015	1 Oct 2015 - 30 Jan 2016

C. Baseline Efficient Frontier Construction

To test the hypothesis that resilience-optimized portfolios offer higher risk-adjusted returns, baseline efficient frontiers for Sharpe-optimized portfolios were first constructed. Historical data was used to find the risks and returns for given portfolio weights. A Monte Carlo simulation was run on this to see the risks and returns of various portfolios with randomly selected weights. For each simulated portfolio, the Sharpe ratio was calculated, indicating the risk-adjusted returns. These risks and returns were plotted onto a scatterplot, and the efficient frontier was plotted by identifying the curve representing the portfolios with the highest returns and lowest risk. The plot also marks the portfolios with the minimum risk and the highest Sharpe ratio.

D. Correlation Analysis

Correlation matrices between assets were calculated for all the indices before, during, and after the black swan events. A breakdown in correlation was observed, which was quantified by using the vector magnitudes of the difference in correlation before and during the event. This provided insights into how assets moved together during the sudden shocks. The impact on diversification and sectoral correlation patterns was analyzed. The correlation difference vector magnitudes were then used as a feature in building the Resilience Index.

E. Resilience Index Construction

A Resilience Index was newly defined and constructed for multidimensional analysis for both risk and recovery. The index utilizes five features, each of which capture different aspects of resilience:

- **Correlation difference vector magnitude** - measures changes in inter-asset relationships
- **Value at risk (VaR) at 99%** - quantifies the tail risk
- **Volatility** - assesses fluctuations in returns.
- **Recovery days** - captures the speed of recovery
- **Recovery ratio** - represents the extent to which prices regained the pre-crisis levels

These features were then normalized using Min-Max scaling to get them within the range of 0 to 1, allowing for comparability. Features that were negatively correlated with resilience were directionally transformed; these transformations were applied to correlation difference, VaR, volatility, and recovery days.

$$f'_i = \begin{cases} 1 - f_i & if\ f_i\ is\ negatively\ correlated\ with\ RI \\ f_i & otherwise \end{cases}$$

where:

- $f_i = feature\ of\ Resilience\ Index$

These features were combined as a weighted sum, using weights which were optimized using heuristics and economic judgement.

$$RI = \sum_{i=1}^{5} w_i f'_i$$

where:

- $\sum_{i=1}^{5} w_i = 1$
- $f_1 = normalized\ correlation\ difference\ vector\ magnitude$
- $f_2 = normalized\ Value\ at\ Risk$
- $f_3 = normalized\ volatility$
- $f_4 = normalized\ recovery\ days$
- $f_5 = normalized\ recovery\ ratio$

Values of the Resilience Index closer to 1 indicate lower correlation breakdown, lower tail risk, faster and fuller recovery, and reduced volatility during crises. The index was computed for both COVID-19 and demonetization to identify the resilient and vulnerable sectors under each circumstance. The Resilience Index values were then integrated into the portfolio optimization, enabling the construction of resilience-optimized efficient frontiers.

F. Sectoral Resilience Comparison

The Resilience Index, both during COVID-19 and during demonetization, was computed for sectoral indices including FMCG, Infrastructure, Pharma, Auto, Media, Realty, Energy, Public Sector Enterprises, and PSU Bank. These values were plotted as a bar graph to compare the relative resilience during both events.

G. Event Agnostic Resilience Index

To achieve transferability and generalization, an event-agnostic combined resilience score was generated. Features like recovery ratio and recovery days were dependent on the event and the policy measures taken to address the event. These features were excluded, and features like correlation difference magnitude, volatility, and VaR were retained. The combined index was calculated by first building a resilience score using the weighted sum of the retained features for both COVID-19 and demonetization. They were further combined using the multiplicative method, where:

$$Combined\ Resilience = \sqrt{RI_{COVID-19} \times RI_{Demonetization}}$$

The multiplicative method was chosen over the additive method to identify assets that were resilient across both the black swan events. This avoids overestimating the resilience of an asset if only one event has been handled well. A high combined Resilience Index indicates that an asset has high resilience across multiple events.

H. Portfolio Optimization with Resilience Index

Efficient frontiers were generated one more time by using resilience-optimized portfolios identified by combining the two metrics: the Sharpe ratio and the Resilience Index.

$$max \;\; F = w_1 \bullet SR + w_2 \bullet RI$$

where:

- $SR = normalized\ Sharpe\ ratio\ of\ the\ portfolios$
- $RI = normalized\ Resilience\ Index\ of\ the\ portfolios$
- $w_1\ and\ w_2\ are\ the\ weights\ assigned\ to\ balance\ between\ efficiency\ and\ resilience$
- $w_1 + w_2 = 1$
- $w_1, w_2 \geq 0$

The table below summarizes how the weights were distributed between the Sharpe ratio and Resilience Index for the different phases of the black swan event.

Phase	Resilience Index	Sharpe Ratio
Before Event	20%	80%
During Event	100%	0%
After Event	60%	40%

The efficient frontiers generated using this weighting scheme were compared with baseline efficient frontiers that were built exclusively using Sharpe ratios from step two. These weights were modelled to have low resilience before black swan periods, high resilience during black swan periods, and medium resilience while the recovery was underway. These weights indicate that investors should shift from return-efficiency before a crisis to resilience during crises, then strike a balance during recovery. This approach identified portfolios with high risk-adjusted returns across all phases.

I. Back Testing on the Chinese Market Crash

The combined Resilience Index was then applied to the Chinese market crash in 2015. The efficient frontiers before, during, and after the Chinese market crash were generated. The efficient frontiers of the resilience optimized portfolios were compared with the baseline efficient frontiers that only focused on Sharpe ratio optimization.

J. Further Out-Of-Sample Tests

Tests were conducted on other black swan events like the global, high-impact, prolonged Russia-Ukraine war in 2022, and local, medium-impact, policy-driven Goods and Services tax (GST) rollout in India in 2017. While the GST rollout was a foreseeable event, its market impact mirrored several characteristics typical of a black swan event. The Russia-Ukraine war was analyzed with 6-month windows for each phase due to its sustained impact on the market. Windows of 3-months for each phase were utilized for the GST rollout due to its short-term impact.

K. Sensitivity Analysis

To test the robustness of the Resilience Index, the event windows were redefined using the Indian Volatility Index (India VIX) by using 30-day moving averages. The during window was from the event date to one month after the peak, the after window was until one month after the volatility stabilized either at pre-event or new normal levels, and the before window matched the length of the after window for symmetry. One month was added to the windows to account for India VIX's 30-day forward looking nature.

IV. RESULTS

A. Sectoral Resilience Index Comparison: COVID-19 vs Demonetization

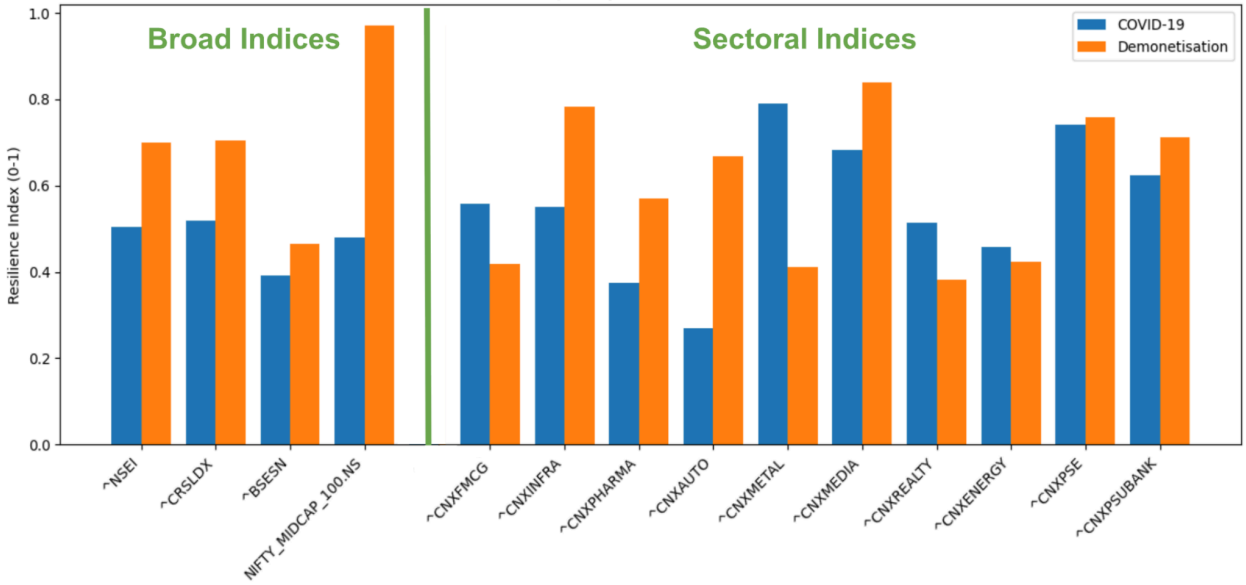


Figure 1: Sectoral Resilience Index Comparison: COVID-19 vs Demonetization

COVID-19 was a global pandemic with a systemic impact across the market. Demonetization, on the other hand, was a local black swan event. As seen in Figure 1, broad indices like Nifty 50, Nifty 500, BSE Sensex, and Nifty Midcap 100 demonstrated a common trend of higher resilience during demonetization compared to COVID-19. The full mapping of ticker symbols to index names is provided in Appendix A.

During COVID-19, Metal was found to be the most resilient, with a Resilience Index of 0.79, while Pharma had the lowest Resilience Index of 0.37, indicating significant instability. On the other hand, during demonetization, the Media was the most resilient sector, with a Resilience Index of 0.84, while the real estate sector was the most negatively affected sector, with a Resilience Index of 0.38.

B. Analysis of Efficient Frontiers Optimized for Sharpe Ratio For COVID-19

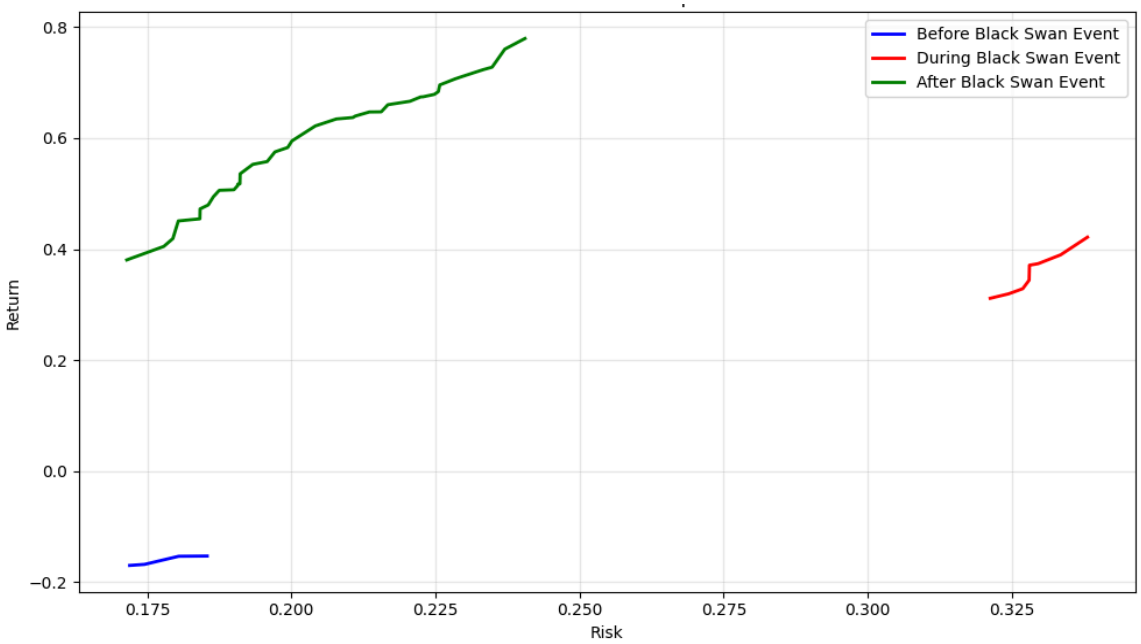


Figure 2: Efficient Frontiers Optimized for Sharpe Ratio for COVID-19

The short-term impact of COVID-19 was assessed by generating the efficient frontiers for a 6-month window before, during, and after COVID-19. These windows were centered around the official announcement date from the World Health Organization (WHO). The efficient frontiers constructed during these periods are presented in Figure 2.

Figure 2 indicates that before the pandemic, there was a low risk and negative or near-zero returns, leading to poor Sharpe ratios. These results reflect the market complacency and underperformance right before the shock. During COVID-19, however, high-risk with moderate returns are evident, suggesting that the risk-return trade-off has worsened. After the event, a wider range of portfolios is visible, with significantly higher returns for lower risk levels, depicting improved Sharpe ratios in the post-event recovery period.

C. Influence of Resilience Index on Efficient Frontiers For COVID-19

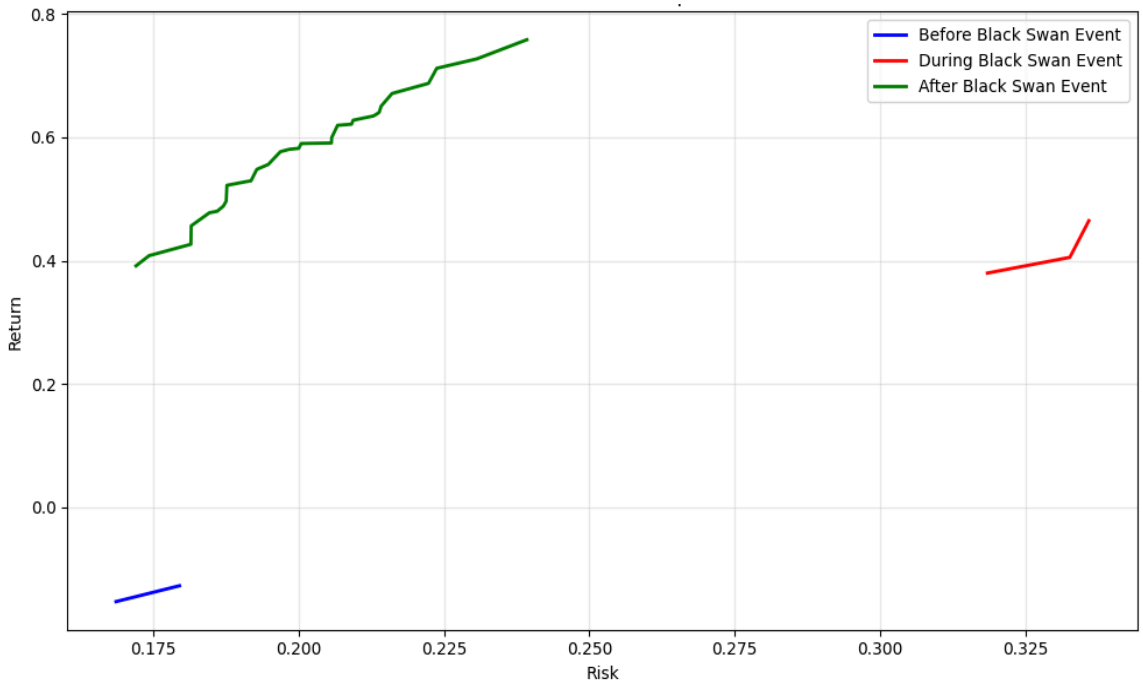


Figure 3: Efficient Frontiers Optimized for Sharpe Ratio and Resilience Index for COVID-19

A differential weighting scheme was used to combine Sharpe ratios and Resilience Index to generate efficient frontiers using resilience-optimized portfolios. Figure 3 shows that returns have increased for the same level of risk during COVID-19, indicating higher efficiency. Notably, there is a similar performance before and after the event, suggesting that the inclusion of resilience does not adversely impact the efficient frontiers in normal conditions.

D. Analysis of Efficient Frontiers Optimized for Sharpe Ratio for Demonetization

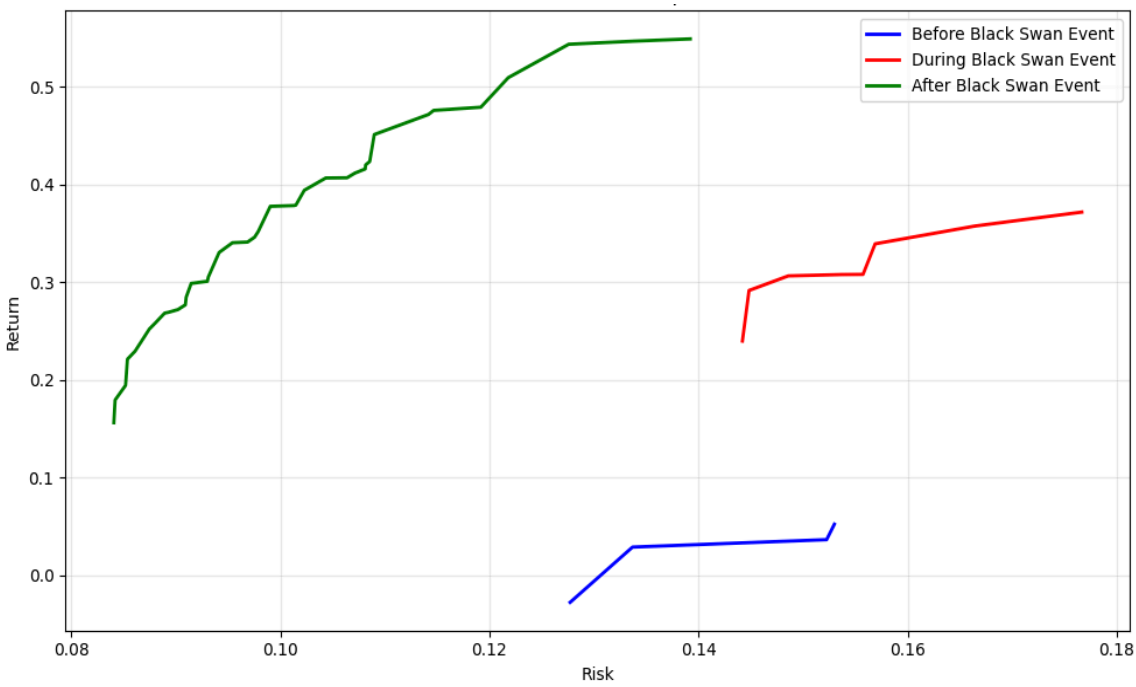


Figure 4: Efficient Frontiers Optimized for Sharpe Ratio for Demonetization

The changes in efficient frontiers for the 2016 demonetization are visualized by utilizing 3-month windows before, during, and after the shock. These windows are centered around the day that the demonetization was announced by the Indian government. The three efficient frontiers are presented in Figure 4.

Figure 4 indicates that before the demonetization, there was a high risk for lower returns, indicating weaker portfolio efficiency before the event. The efficient frontier during demonetization is characterized by high risk with high returns, depicting improvements in the risk-adjusted returns. After the shock, a wider range of portfolios is visible, with lower overall risk and significantly higher returns, depicting stable, high-performing asset opportunities.

E. Influence of Resilience Index on Efficient Frontiers for Demonetization

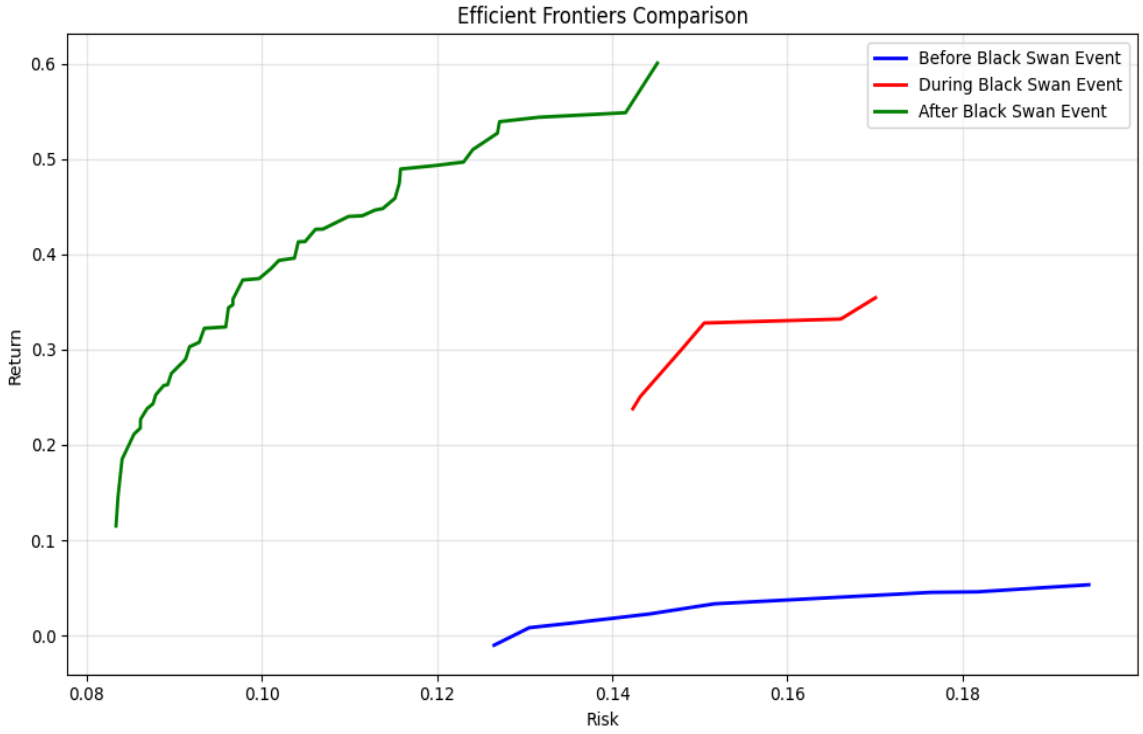


Figure 5: Efficient Frontiers Optimized for Sharpe Ratio and Resilience Index for Demonetization

Figure 5 depicts an increase in return levels for similar risk levels before the demonetization. During and after the event, portfolios exhibit a lower risk level for similar returns. Hence, the Resilience Index has improved the efficiency of the portfolios either by increasing returns or reducing risk in all phases.

F. Generalization of Resilience Index

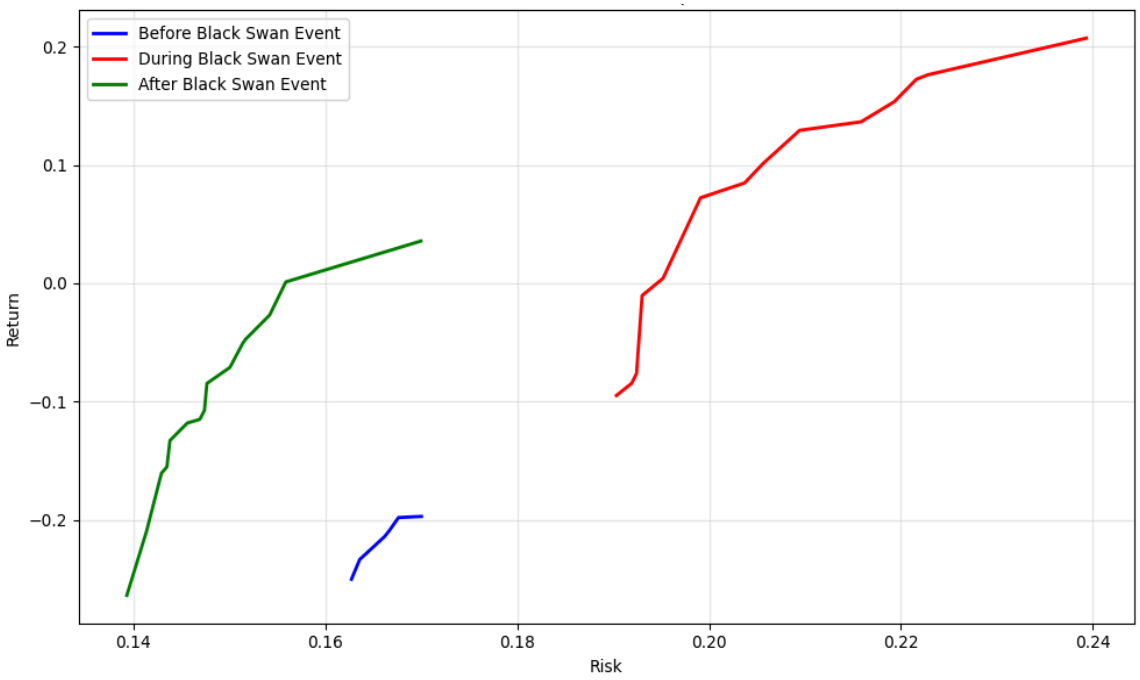


Figure 6: Efficient Frontiers Optimized for Sharpe Ratio for the Chinese Market Crash

The Resilience Index was tested for its generalizability. Event-agnostic features of the Resilience Index of COVID-19 and that of demonetization were combined using the geometric mean. This combined Resilience Index was used to generate efficient frontiers for a third event, the Chinese market crash.

Figure 6 illustrates deeply negative returns for low risk before the Chinese market crash. During the crash, there are higher returns but with increased risk levels, indicating low stability. After the event, on the other hand, there are higher returns for lower risk levels, pointing to high Sharpe ratios.

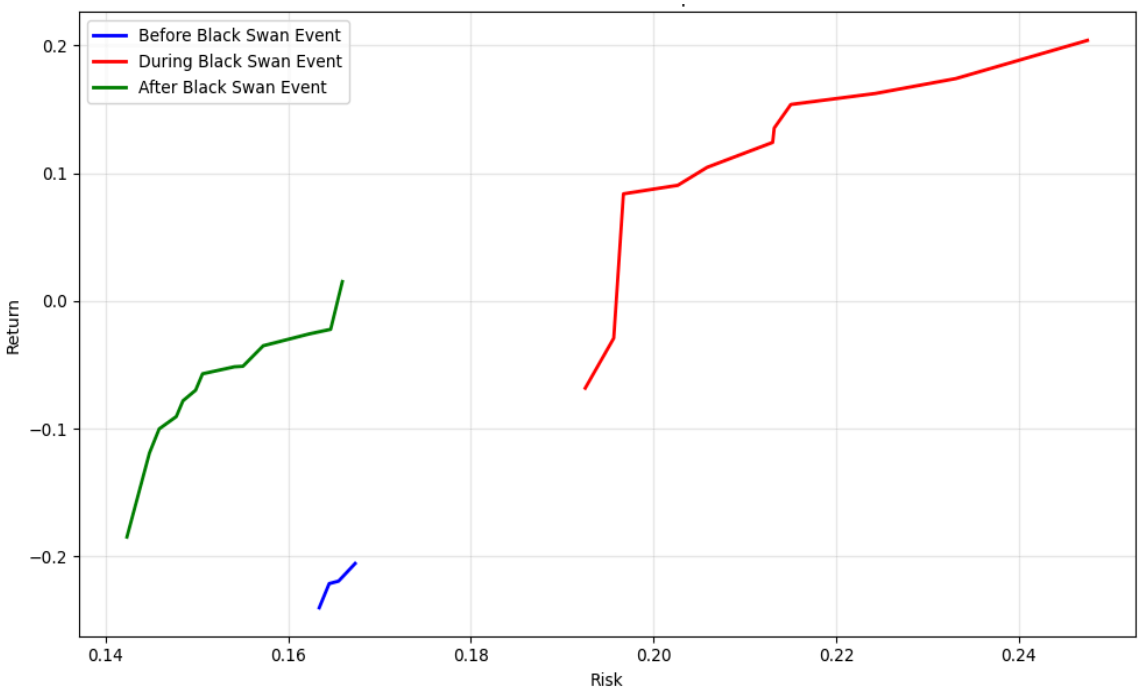


Figure 7: Efficient Frontiers Optimized for Sharpe Ratio and Resilience Index for Chinese Market Crash

Figure 7 depicts a similar graph before the crash. However, during the event, portfolios have higher returns, indicating better risk-adjusted returns. After the crash, the efficient frontier has positive or near-zero returns, suggesting a faster recovery.

G. Additional Tests

Out-of-sample tests for the Russia-Ukraine war and the Goods and Services Tax (GST) rollout were conducted. For the Russia–Ukraine war, 6-month event windows centered on the announcement date were used. For the GST rollout, 3-month event windows centered on the rollout date were analyzed. The trends observed during these events are consistent with the observations from the three black swan events listed in the Results section (B-F). The results for the Russia-Ukraine war indicate that portfolios optimized for resilience had lower risk levels

before the event and higher returns in both the during and after time frames, pointing to better risk-adjusted returns in all three phases. Results for the GST rollout show that resilient portfolios had higher returns in all three phases of the event, indicating stronger Sharpe ratios.

Sensitivity analysis was conducted with event windows informed by the India Volatility Index (India VIX) for all tested events. Upon finding new efficient frontiers for these windows, it was seen that the results were consistent with the findings, where efficient frontiers constructed using the Resilience Index had higher risk-adjusted returns.

V. DISCUSSION

A. Interpretation Of Results

The newly introduced Resilience Index was used to conduct a sectoral analysis to identify the most resilient sector in the Indian stock market during COVID-19 and demonetization.

As seen in Figure 1, Nifty Midcap 100 demonstrated exceptional resilience during demonetization, as it benefited from the digitization and formalization of the economy. Reduced exposure of foreign institutional investors on the midcap index compared to large caps allowed it to avoid the impacts of selling and benefit from the confidence spike from domestic investors (Kumara, N. V., 2017). Moreover, Figure 1 shows that during demonetization, the Media sector displayed the highest resilience. This is because it experienced a compound annual growth rate (CAGR) of 9.1% from 2016 to 2021, indicating a strong recovery after the demonetization (KPMG India, 2017). Figure 1 further shows that Realty had the lowest resilience. This is because real estate transactions in India primarily depend on cash transactions, specifically in Tier 2 and Tier 3 cities. A 40% dip in property registrations was witnessed immediately after

demonetization. The recovery of the real estate sector was prolonged and continued to happen even in 2024 (Capital, T., & Capital, T., 2024).

During COVID-19, Metals were found to be the most resilient sector. Metals like copper hit a five-year high in October 2021, showing a strong resilience and ability to recover after COVID-19 (Datta, D, 2021). Pharmaceuticals had the lowest resilience due to supply chain disruptions of Active Pharmaceutical Ingredients (APIs) imported from China, leading to supply shortages and margin pressures. (dst.gov.in, 2020)

While sector-level resilience varied across events, the broader question was whether these insights could translate into stronger, crisis-proof portfolios. Given that different sectors have different levels of resilience to black swan events, as indicated in Figure 1, diversified portfolios that use assets filtered by the Resilience Index can outperform standard portfolio construction methods. These methods include minimum variance optimization and Sharpe-only optimization. This study validated this hypothesis by constructing efficient frontiers for COVID-19 and demonetization. These were first constructed by optimizing only for Sharpe ratios, and then for a combination of Sharpe ratios and Resilience Index. The study used time windows centered around the black swan event to study the short-term impact.

The black swan events were then split into three different phases: before, during, and after. Resilience was differently weighted in the multi-objective function involving the Sharpe ratio and Resilience Index, depending on the phase. For example, a 20% weightage was given to resilience before the event, 100% weightage during the event, and 60% weightage after the event in the recovery phase. The differential weighting scheme performed better than equal weightage

for both the Sharpe ratio and Resilience Index. These findings suggest that dynamically adjusting weights based on market indicators could further enhance performance.

Sharpe-optimized portfolios do well in normal conditions but underperform during black swan events. For example, the efficient frontiers before COVID-19 are quite similar, but the efficient frontier during the pandemic has higher returns with resilience than without. Hence, it can be concluded that Sharpe-only portfolios fail to capture deep downside risk or recovery strength, justifying the need for the inclusion of the Resilience Index in portfolio optimization. Therefore, combining both the Resilience Index and the Sharpe ratio optimizes for overall performance, as the Sharpe ratio ensures higher risk-adjusted returns and the Resilience Index allows for risk mitigation during extreme events.

It is observed that this combined Resilience Index can outperform Sharpe-optimized portfolios in all phases. The consistent performance improvement for the Chinese market crash (Figure 7), the Russia-Ukraine war, and the GST rollout suggests that the Resilience Index can be generalized and utilized for other black swan events. Its robustness is further reinforced by the sensitivity analysis using volatility-driven event windows.

Overall, the results depict that in all the events, portfolios optimized using the Resilience Index showed a shift in the efficient frontier leftwards or upwards. For example, after the demonetization, there was a 3% increase in returns due to the use of the Resilience Index.

B. Limitations

While the Resilience Index is seen to be effective in creating resilient portfolios that outperform during black swan events, several limitations of the model must be acknowledged.

Firstly, factor weights were specifically selected for COVID-19 and demonetization. The effectiveness of this selection for other types of black swan events is unknown. Moreover, the index depends on historical data, assuming that past patterns of risk and returns are good indicators of future trends. Additionally, the features considered, including volatility, correlation difference magnitude, VaR, recovery ratio, and recovery rate, are justifiable but not exhaustive. Other variables, like government intervention and market sentiment, will need to be considered for incorporating the irrational exuberance of the market in the short term. Furthermore, differential weights were effective in capturing the importance of resilience during different phases. These weights will need to be dynamically adjusted as the market evolves. Finally, the Resilience Index was validated for a limited number of black swan events. The generalized version of the Resilience Index did not consider event-specific recovery metrics. An extensive back testing across multiple black swan events from the inception of the stock market will need to be conducted to establish holistic generalizability.

C. Comparison to Existing Literature

To position the Resilience Index within the context of existing financial risk-related research, this section reviews other complementary approaches to measuring risk and managing portfolio stability during market disruptions. Prior studies focus on tail risk measures and investor sentiment – factors often overlooked by traditional risk metrics.

Georg Mainik, Georgi Mitov, and Ludger Rüschendorf (2015) studied the use of tail risk measures to optimize portfolios during rare events. This study defined a new metric called the Extreme Risk Index (ERI), which was computed using multivariate extreme value theory. Upon back testing on the S&P 500, the study found that the ERI strategy significantly outperformed traditional metrics with 11.5% annualized returns, particularly when dealing with stocks with heavy-tailed distributions. The Resilience Index can use ERI as another factor specialized in capturing tail risks during black swan events, similar to VaR.

Jyothi Singhal (2023) investigated typical investor behaviors and the role they play in investment decision-making. The paper discusses the calculation of a Fear and Greed Index as a quantitative measure to track investor sentiments. This index can be combined with the Resilience Index, as it provides a behavioral metric for market movements. While the Resilience Index measures historical market stability, the Fear and Greed Index indicates the possible underlying factors that may have triggered or prolonged the crisis. Therefore, combining both indices could be a useful tool for investors. Investors could gauge the periods where the market is driven by extreme emotions by using the Fear and Greed index, and build a resilient portfolio during such conditions using the Resilience Index. This allows for informed decision-making based on both historical performance and the current psychological state of the market.

D. Future Research

This study can be extended by generating the weights for combining the different features of the Resilience Index using machine learning (regression models). In this approach, weights can be learned by predicting the shift in risk and returns using the contributing factors. Due to

insufficient training data from rare black swan events, synthetic data generation will be required. Similarly, weights for combining the Sharpe ratio with the Resilience Index for portfolio optimization can be learned using machine learning. These weights can be dynamically updated by using market indicators like customer sentiment, policy interventions, and economic data. Researchers may also explore ways to predict recovery metrics for new events based on existing events and use new features in addition to the ones used in this study to compute the Resilience Index.

VI. CONCLUSION

This study proposes a Resilience Index that uses a weighted combination of correlation difference magnitude, volatility, VaR, recovery ratio, and recovery days as a practical metric to assess asset stability and performance during black swan events. Applied to COVID-19 and the Indian demonetization, the index revealed clear sectoral resilience patterns, emphasizing the relevance of resilience in portfolio construction. When combined with the Sharpe ratio using a phase-specific differential weighting scheme, the Resilience Index improved or maintained the same risk-return ratio during the short-term analysis of all the black swan events. Its generalizability was further evident through the application to the Chinese market crash, implying that utilizing the Resilience Index in asset selection for portfolios offers higher stability and stronger risk-adjusted returns. Overall, this study highlights the potential of resilience metrics in rebalancing portfolios during and after disruptive events, providing investors with a systematic defense against market shocks.

DATA AND CODE AVAILABILITY

The complete source code for reproducing this study's results are available at: https://github.com/ananya222/resilience-index under the MIT license.

REFERENCES

Flage, R., & Aven, T. (2015). Emerging risk–Conceptual definition and a relation to black swan type of events. Reliability Engineering & System Safety, 144, 61-67.

Aven, T. (2013). On the meaning of a black swan in a risk context. Safety science, 57, 44-51.

Walter, C. (2020). Financial Black Swans: unpredictable threat or descriptive illusion?. In Societies Under Threat: A Pluri-Disciplinary Approach (pp. 173-186). Cham: Springer International Publishing.

Baker, S. R., Bloom, N., Davis, S. J., Kost, K., Sammon, M., & Viratyosin, T. (2020). The unprecedented stock market reaction to COVID-19. The review of asset pricing studies, 10(4), 742-758.

Ozili, P. K., & Arun, T. (2023). Spillover of COVID-19: impact on the Global Economy. In Managing inflation and supply chain disruptions in the global economy (pp. 41-61). IGI Global Scientific Publishing.

Singh, M. K., & Neog, Y. (2020). Contagion effect of COVID‐19 outbreak: Another recipe for disaster on Indian economy. Journal of Public Affairs, 20(4), e2171.

Donadelli, M., Kizys, R., & Riedel, M. (2017). Dangerous infectious diseases: Bad news for Main Street, good news for Wall Street?. Journal of Financial Markets, 35, 84–103. https://doi.org/10.1016/j.finmar.2016.12.003

Aslam, F., Mohmand, Y. T., Ferreira, P., Memon, B. A., Khan, M., & Khan, M. (2020). Network analysis of global stock markets at the beginning of the coronavirus disease (Covid-19) outbreak. Borsa Istanbul Review, 20, S49-S61.

Kumara, N. V. (2017). Effectiveness of FII on the performance of NSE indices. International Journal of Management, IT and Engineering, 7(2), 183-192. https://www.indianjournals.com/article/ijmie-7-2-015

Datta, D. (2021, October 18). Metal stocks: Bulls have the upper hand, Copper at a five-year high. https://www.business-standard.com/article/companies/metals-bulls-seem-to-have-an-upper-hand-over-bears-121101801245_1.html

TIFAC releases report on Active Pharmaceutical Ingredients- Status, Issues, Technology Readiness and Challenges | Department of Science & Technology. (n.d.).

https://dst.gov.in/tifac-releases-report-active-pharmaceutical-ingredients-status-issues-technology-readiness-and

KPMG India. (2017, April). FICCI Frames 2017 (2017). KPMG India. Retrieved from https://assets.kpmg.com/content/dam/kpmg/in/pdf/2017/04/FICCI-Frames-2017.pdf

Capital, T., & Capital, T. (2024, July 19). The impact of demonetization on real estate in India. TATA Capital Blog. https://www.tatacapital.com/blog/loan-for-home/impact-of-demonetization-on-real-estate-in-india/

Shankar, R., & Dubey, P. (Eds.). (2021). Indian stock market during the COVID-19 pandemic: Vulnerable or resilient?: Sectoral analysis. Vilnius University Press. https://doi.org/10.15388/omee.2021.12.51

Dharmapala, D. and Khanna, V.S. (2019), Stock Market Reactions to India's 2016 Demonetization. Journal of Empirical Legal Studies, 16: 281-317. https://doi.org/10.1111/jels.12213

Sivankutty, N. S. (2017). Impact of demonetization on gross domestic product (GDP) – An analytical approach. In the International Conference on Emerging Trends in Engineering, Science and Management (pp. 1399–1404).

Rezaei Sou, H., Esfahanipour, A., & Akbarpour Shirazi, M. (Eds.). (2021). A quantitative measure of financial resilience of rms: Evidence from Tehran stock exchange. Scientia Iranica E. https://scientiairanica.sharif.edu/article_22141_087244720f493f453af4d06e43bda81f.pdf

Sureshkumar, V., & Balasubramanian, P. (2021). Risk and returns of Indian listed firms after demonetization. Cogent Economics & Finance, 9(1). https://doi.org/10.1080/23322039.2021.1920688

Anoop, P., Parab, N., & Reddy, Y. V. (2018). Analyzing the impact of demonetization on the Indian stock market: Sectoral evidence using the GARCH model. Australasian Accounting, Business and Finance Journal, 12(2), 104–116. https://doi.org/10.14453/aabfj.v12i2.7

Sahil, S. K., Dhananjhay, G., & Karim, S. (Eds.). (2024). A Study on the Impact of Black Swan Events on the Indian Stock Market. International Journal of Research in Engineering. https://indusedu.org/pdfs/IJREISS/IJREISS_4479_62352.pdf

Kumar, G. P., & Pavani, G. (2024). Black Swan Events and Their Impact on Indian Financial Markets with Reference to Nifty50 Index. The Review of Contemporary Scientific and Academic Studies, 4(12). https://doi.org/10.55454/rcsas.4.12.2024.001

Mainik, G., Mitov, G., & Rüschendorf, L. (2015). Portfolio optimization for heavy-tailed assets: Extreme risk index vs. Markowitz. Journal of Empirical Finance, 32, 115–134. https://doi.org/10.1016/j.jempfin.2015.03.003

Singhal, J. (2023, October 27). Role of greed and fear index in investment decision making. International Conference on Leveraging Technology for Creating Competitive Advantage. SSRN. https://doi.org/10.2139/ssrn.4614883

APPENDIX

Index and Ticker Mapping

Ticker Symbol	Index Name	Type
^NSEI	Nifty 50	Broad Market
^CRSLDX	Nifty 500	Broad Market
^BSESN	BSE Sensex	Broad Market
^NSEMDCP100	Nifty Midcap 100	Broad Market
^CNXFMCG	Nifty Fast Moving Consumer Goods	Sectoral Index
^CNXINFRA	Nifty Infrastructure	Sectoral Index
^CNXPHARMA	Nifty Pharma	Sectoral Index
^CNXAUTO	Nifty Auto	Sectoral Index
^CNXMETAL	Nifty Metal	Sectoral Index
^CNXMEDIA	Nifty Media	Sectoral Index
^CNXREALTY	Nifty Realty	Sectoral Index
^CNXENERGY	Nifty Energy	Sectoral Index
^CNXPSU	Nifty Public Sector Enterprise	Sectoral Index
^CNXPSUBANK	Nifty Public Sector Undertaking Banks	Sectoral Index